

SMART

WATCH



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TA

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Code

CSC2104

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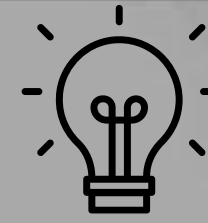
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Project Idea



- The current generation of premium Smart Watches (e.g., Apple Watch, Samsung Galaxy) are often highly expensive, with prices ranging from 10,000 to 20,000 EGP.
- Our project is to develop a specialized Smart Watch that focuses primarily on essential health monitoring—Heart Rate and Steps—using affordable, yet accurate, components.
- This allows us to offer a highly functional, dedicated health tracker at a competitive price point, projected to be around 4,500 EGP, bridging the gap between basic fitness bands and overpriced premium devices.

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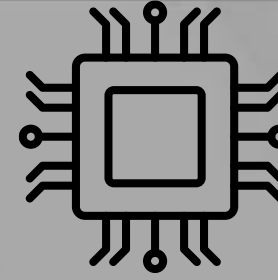
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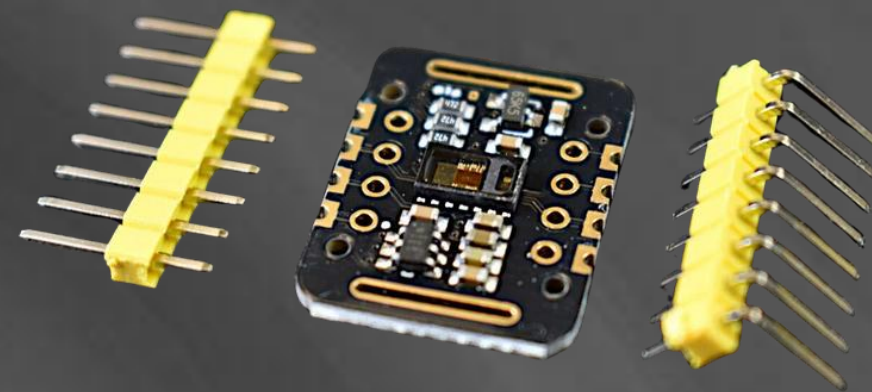


Project Components

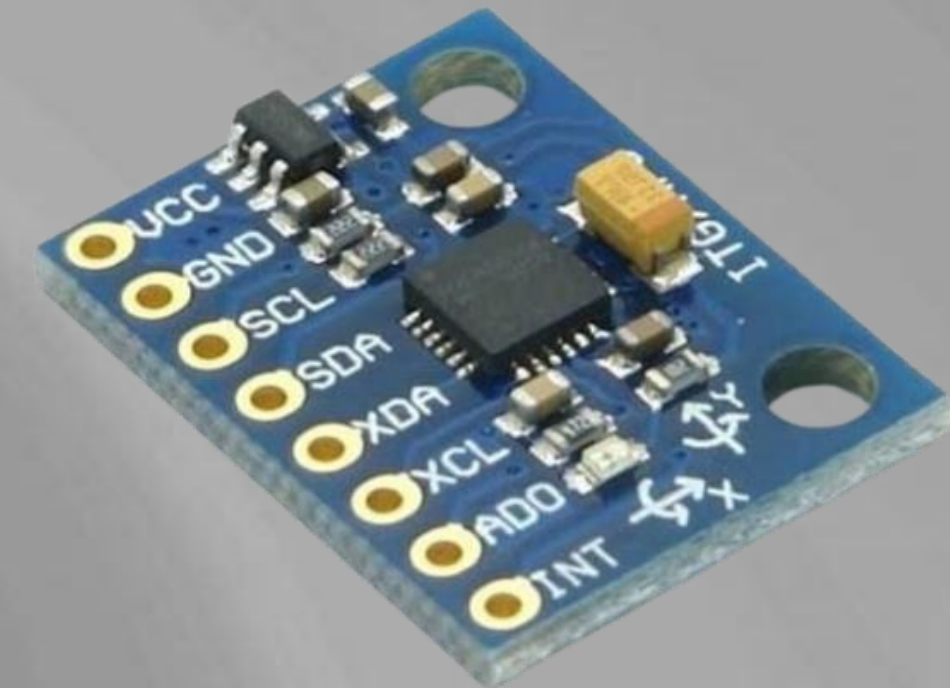
ESP32



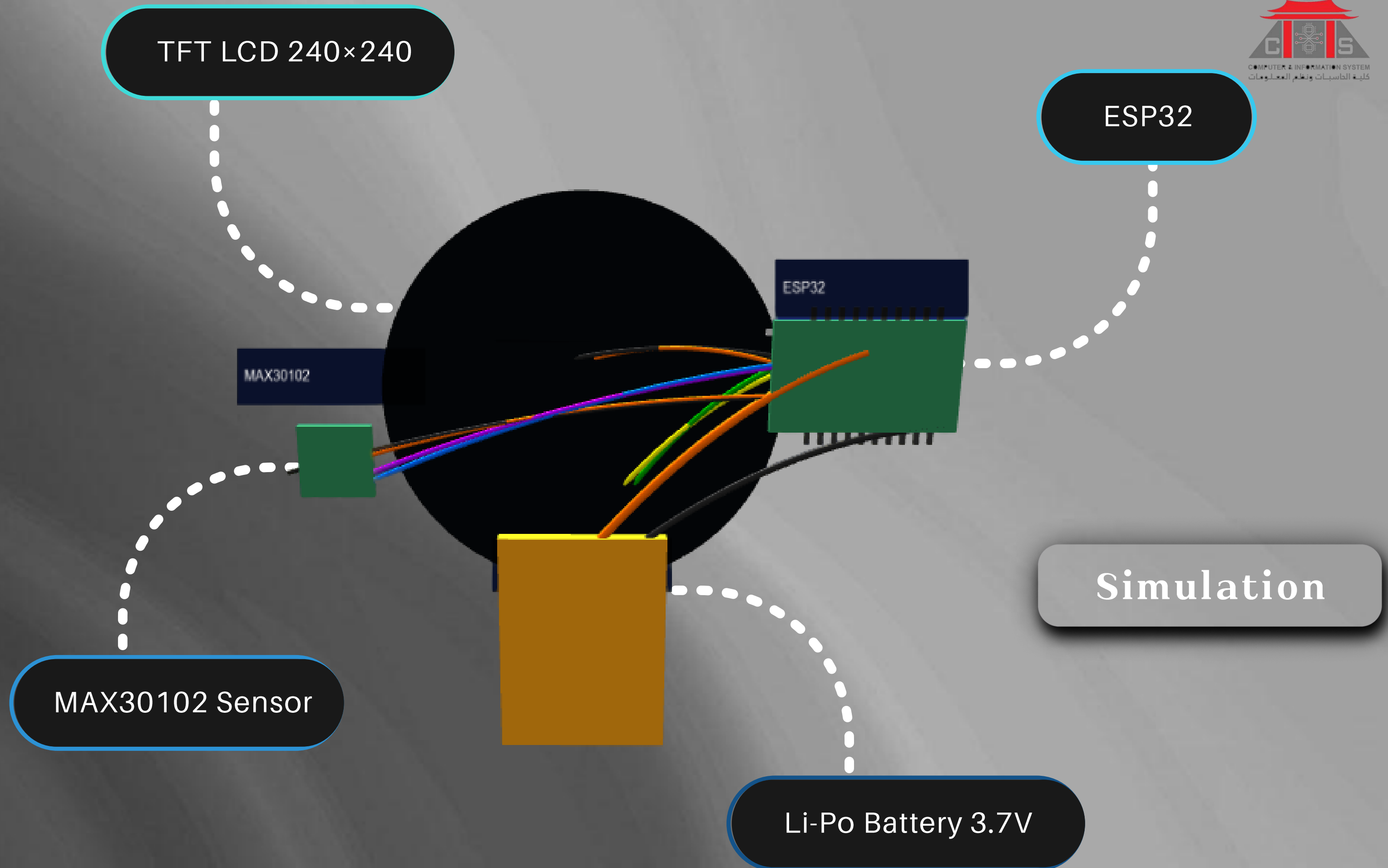
TFT LCD
240×240



MAX30102 Sensor



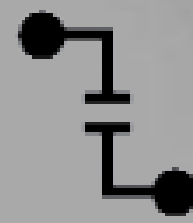
MPU6050 Sensor



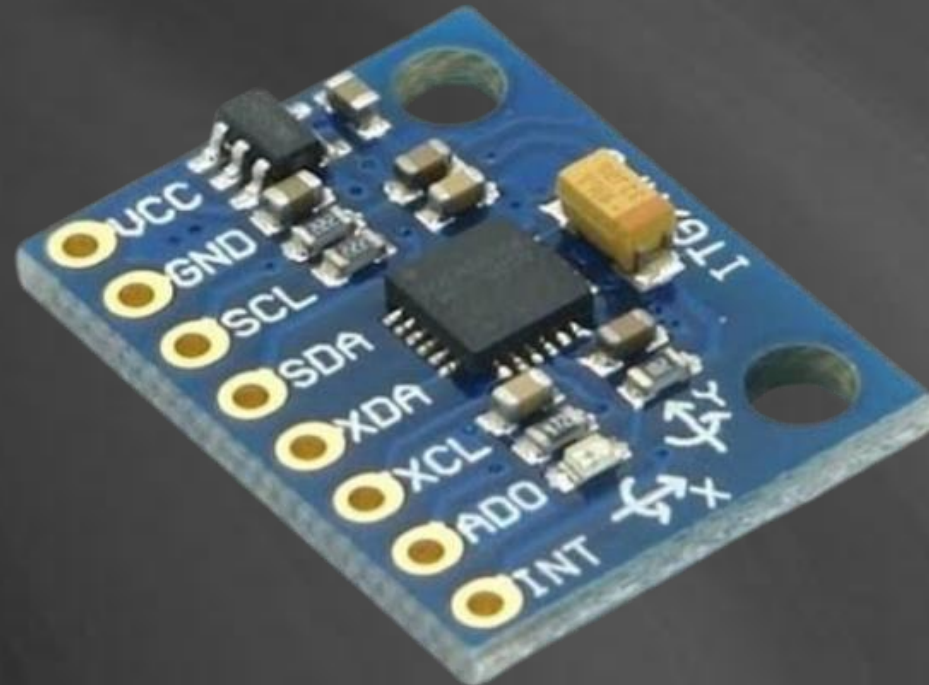
The ESP32 serves as the central processing unit, managing wireless and component communication:

- **Wireless:** Handles Wi-Fi and Bluetooth Low Energy (BLE).
- **Protocols:** Uses SPI for the TFT Display and I²C for the MAX30102 sensor.
- **Key Pins:** SDA=GPIO21, SCL=GPIO22, MOSI=GPIO23, SCK=GPIO18, CS=GPIO5.





Connections



1. The MPU6050 (Motion Sensor) is connected to the ESP32 using the I2C Interface as follows:

MPU6050 (VCC) → ESP32 (3V3)

MPU6050 (GND) → ESP32 (GND)

MPU6050 (SDA) → ESP32 (GPIO 21)

MPU6050 (SCL) → ESP32 (GPIO 22)

MPU6050 (INT) → ESP32 (GPIO 15): (Optional)
Interrupt pin to alert the CPU of motion events.



2. TFT Display Connections (SPI Protocol)

The SPI (Serial Peripheral Interface) protocol is used for high-speed data transfer to the display.

- esp32 → tft: SPI MOSI (GPIO23)
- esp32 → tft: SPI SCLK (GPIO18)
- esp32 → tft: Power (3.3V)
- esp32 → tft: Ground (GND)

Control Pin

- esp32 → tft: Chip Select (CS) = GPIO5

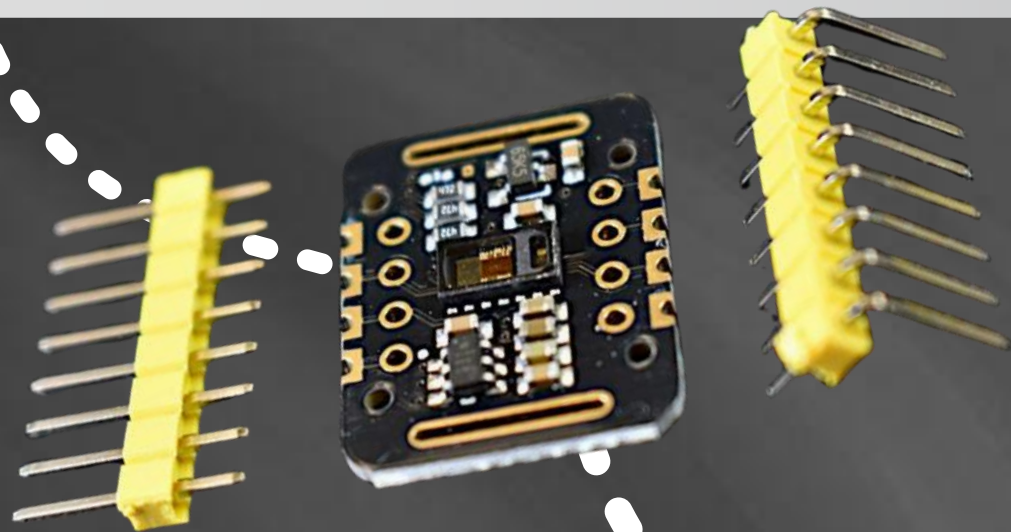
TFT 240×240 Display

- SPI display. Connect MOSI, SCLK, CS, DC, RST. Drive with ILI9xxx libraries. Keep SPI lines short and shielded in wearables.

3. MAX30102 Sensor Connections (I²C Protocol)

The I²C protocol is used to communicate with the heart rate and SpO₂ sensor.

- esp32 → max30102: I²C SDA (GPIO21)
- esp32 → max30102: I²C SCL (GPIO22)
- esp32 → max30102: Power (3.3V)
- esp32 → max30102: Ground (GND)



MAX30102 Sensor

Optical heart-rate & SpO₂ sensor. I²C device; needs good skin contact and signal processing.
Pins: VCC (3.3V), GND, SDA, SCL, INT.

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Code General Overview

"This code functions as a mini-operating system for a smart watch powered by the ESP32 microcontroller. The core philosophy of the code is built on Concurrent Data Processing, where the system simultaneously acquires and analyzes data from motion sensors (MPU6050) and biometric sensors (MAX30102).

Beyond just displaying raw data, the code features a robust Signal Processing Engine that filters out noise and extracts accurate results using algorithms like 'Moving Averages.' It also utilizes a State Machine logic to manage seamless navigation between UI screens, integrated with an internal software RTC to maintain precise timekeeping without external modules. In short, the code focuses on three pillars:

Measurement Accuracy, System Responsiveness, and User Interface Simplicity."

Code General Overview

- **Heart Rate Monitoring (BPM):** Processes raw IR signals to detect heartbeats and calculate a stable average.
- **Oxygen Saturation (SpO2):** Uses the Ratio-of-Ratios algorithm to estimate blood oxygen levels.
- **Smart Pedometer:** A custom step-detection algorithm that filters out random noise and movements.
- **Real-Time Clock (RTC):** A software-based clock to track time (HH:MM:SS) during monitoring.
- **Multi-Screen UI:** Professional interface with three dedicated modes: Heart Rate, Activity, and SpO2.



System General Overview

1. Concurrent Data Processing: The firmware is designed as a Mini-Operating System running on the ESP32 microcontroller. Its core architecture manages a synchronized "data flow" from multiple sensors—the MAX30102 (Biometrics) and the MPU6050 (Motion)—simultaneously. The system doesn't just display raw values; it captures, processes, and analyzes data in real-time before visualization.



System General Overview

- 2. Signal Processing & Advanced Algorithms:** The defining feature of this code is its Signal Processing Engine. Instead of relying on raw sensor output, the data passes through three sophisticated stages:
- **Moving Average Filter:** Smooths out fluctuations in Heart Rate and SpO2 readings to eliminate "jumpy" or erratic values.
 - **Dynamic Calibration:** The motion sensor auto-calibrates every 5 seconds, allowing the system to adapt to the user's movement intensity and distinguish between actual steps and random vibration.
 - **AC/DC Component Filtering:** For oxygen monitoring, the code separates constant light absorption (DC) from pulsing blood flow (AC) to ensure medical-grade calculation accuracy.



System General Overview

- 3. Intelligent UI Management:** To ensure a high-performance user experience, the system utilizes Partial Refresh logic. Instead of redrawing the entire screen—which causes flickering—the code identifies and updates only the specific pixels that have changed. This optimizes CPU usage and ensures a smooth, professional interface.
- 4. Firmware Intelligence & Diagnostics:**
- **Self-Diagnostics (I2C Scanner):** Upon startup, the system performs a hardware scan to verify the integrity of all sensor connections.
 - **Software-based RTC:** A high-precision digital clock is built directly into the firmware using internal timers, maintaining accurate time (HH:MM:SS) without needing an external RTC module.
 - **Finite State Machine (FSM):** The system manages three distinct operational modes (Heart Rate, Activity, SpO2) through a seamless single-button interface.



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Waterproof Design (IP68): Enhancing the hardware casing to be fully water-resistant, allowing for use during swimming and in all weather conditions.

Mobile App Integration: Developing a dedicated smartphone application to sync data and view long-term health trends via Bluetooth Low Energy (BLE).



Thanks and Appreciation :

We would like to sincerely thank Dr. Soha Safwat, Dr. Haitham Farouk Dr. Muhamed Abdulhadi , Eng. Nada Abdelhamid for their valuable support and great efforts during our project.

Your guidance and encouragement meant a lot to us and truly made a difference.

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